

Light: In-class Work

Year 9 Science · about 25 minutes in four parts

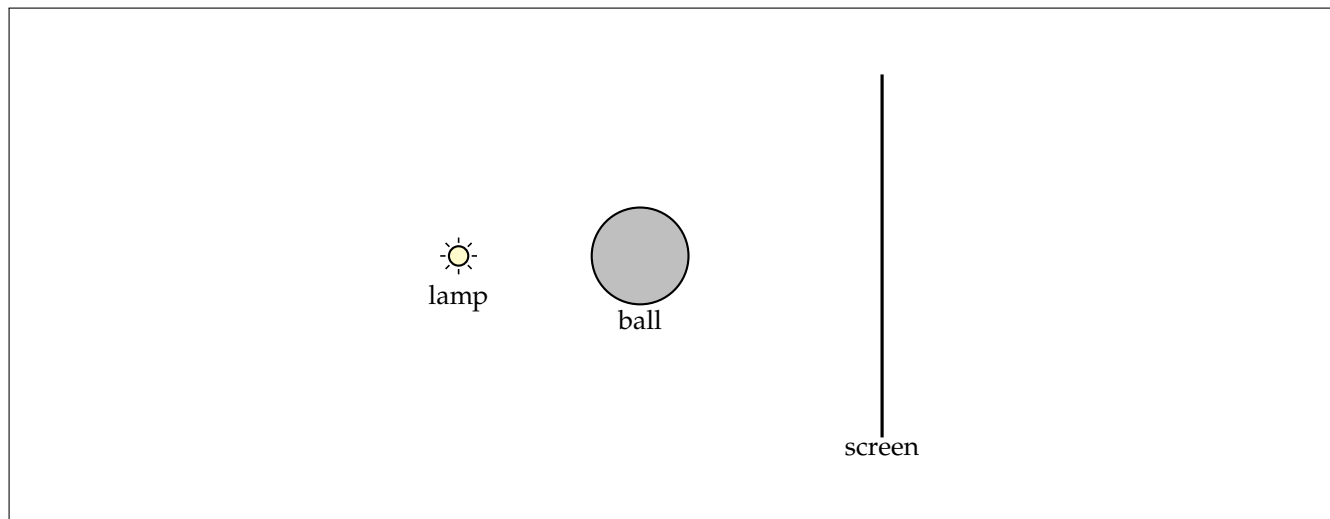
Name: _____

Use a sharp pencil and a ruler for every ray. Put **arrows** on rays and draw virtual rays as **dotted** lines.

Part A Shadows and colour

about 7 minutes

A1 A small lamp shines on a ball in front of a screen. Draw two rays from the lamp that just touch the top and bottom of the ball and reach the screen. Shade the shadow and label the umbra.



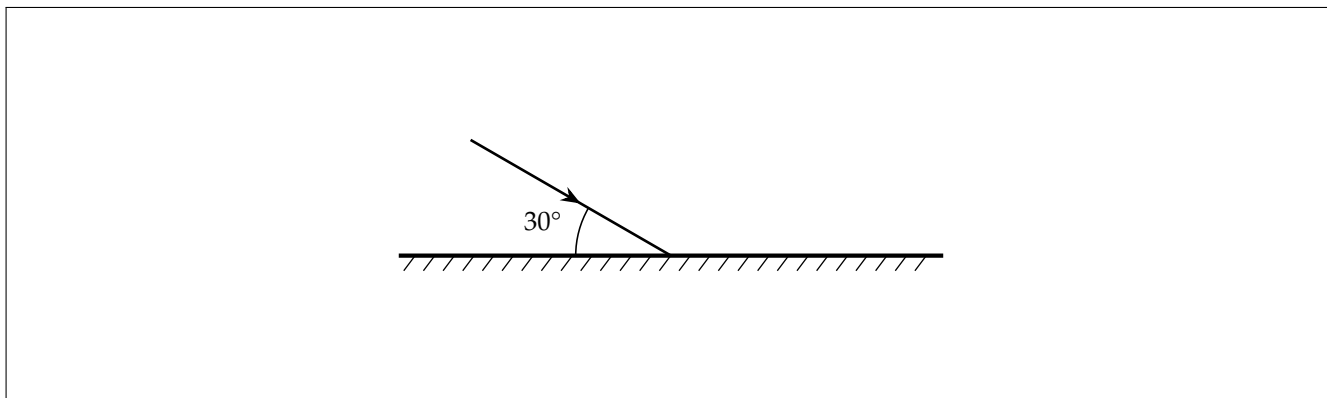
A2 Complete the table.

Object	Light shone on it	Colour it appears
white paper	green light	
blue car	white light	
blue car	red light	
yellow banana	green light	
black shoe	white light	

A3 Complete the colour mixing for light.

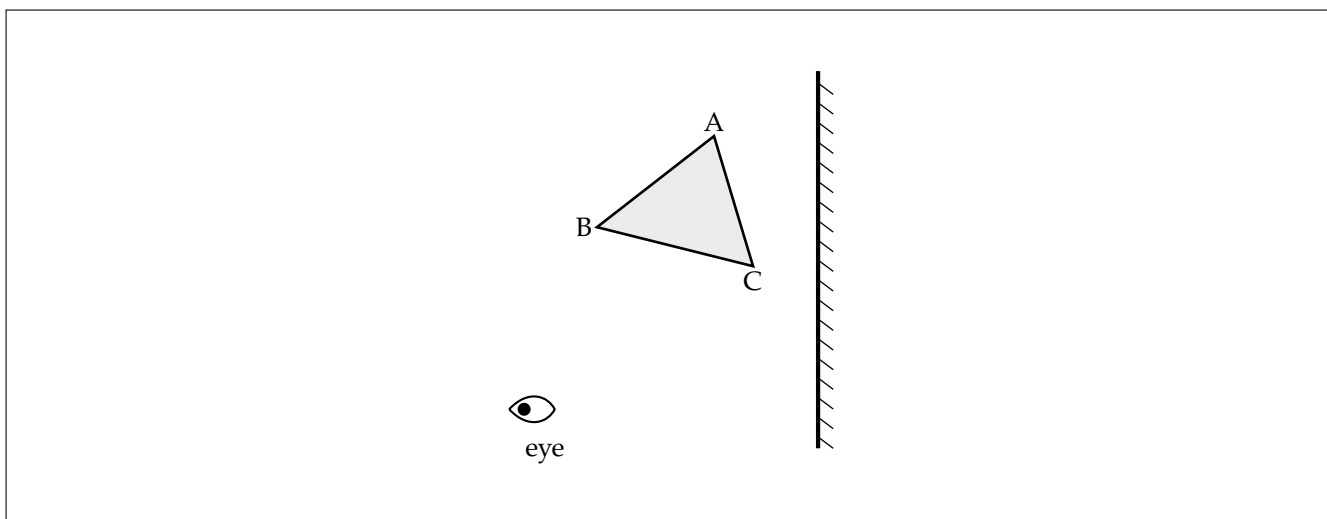
red light + blue light = _____ green light + blue light = _____

B1 A ray hits a plane mirror at 30° to the mirror surface. Draw the normal and the reflected ray. Label the angle of incidence i and the angle of reflection r , and write their sizes.

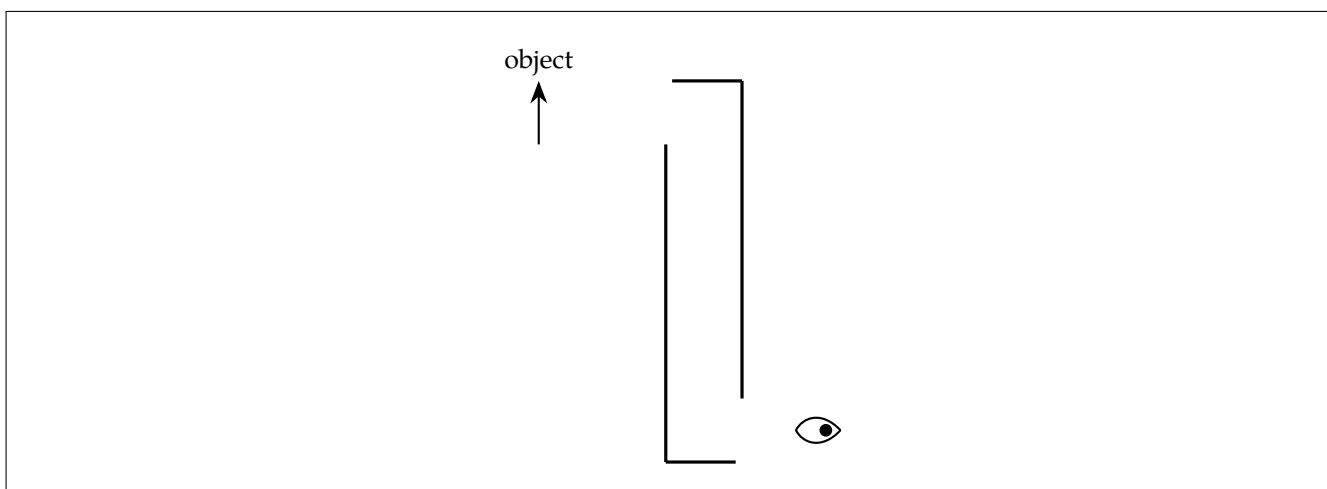


$i =$ _____ $r =$ _____

B2 Draw the image $A'B'C'$ of triangle ABC in the plane mirror. Then draw two rays from A that reflect off the mirror into the eye.



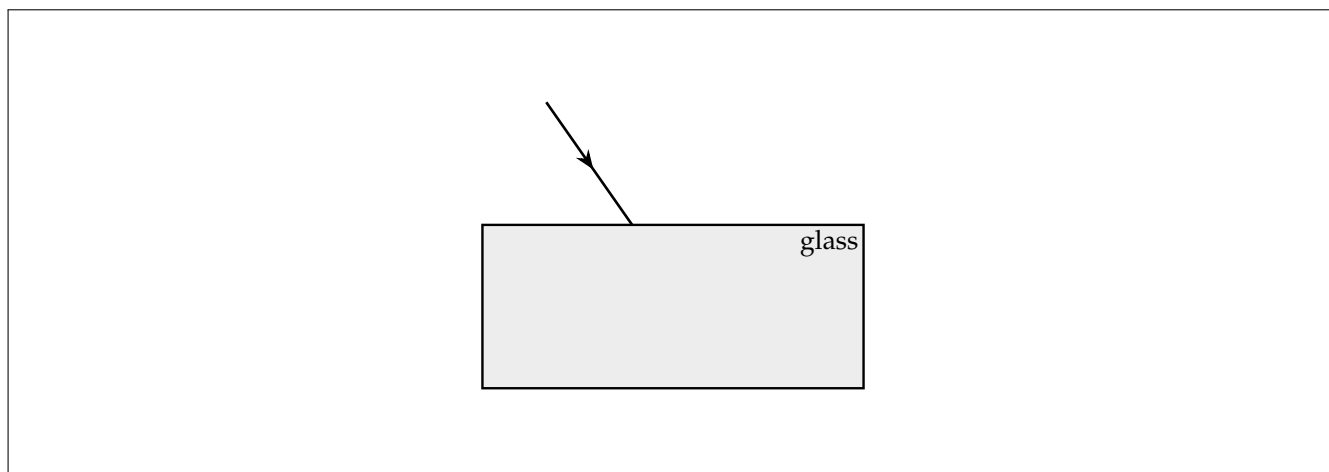
B3 Complete the periscope: draw the two mirrors and the path of a ray from the object to the eye.



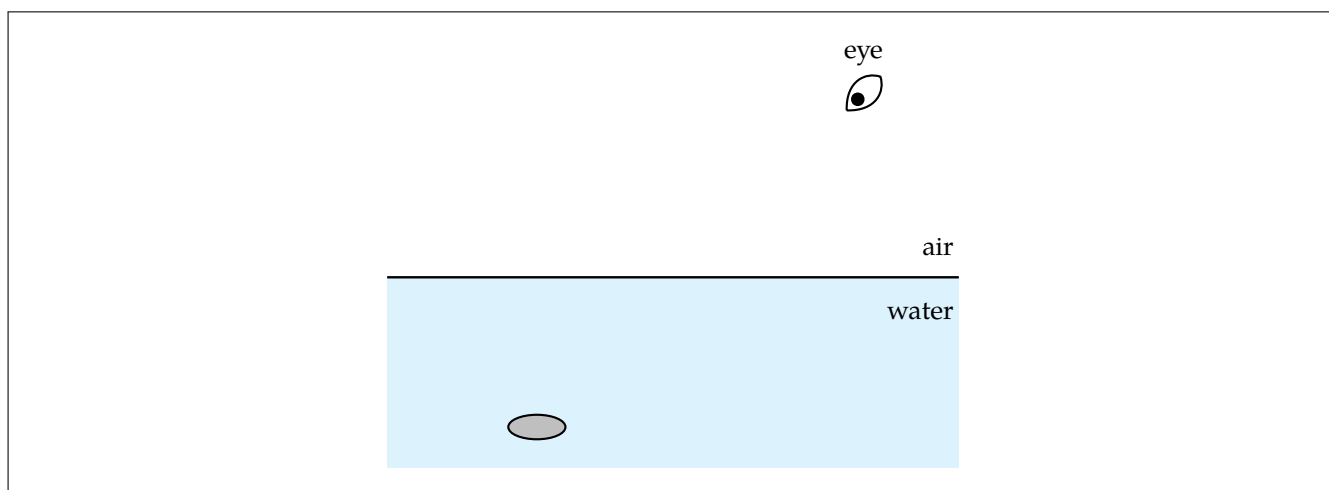
Part C Refraction

about 6 minutes

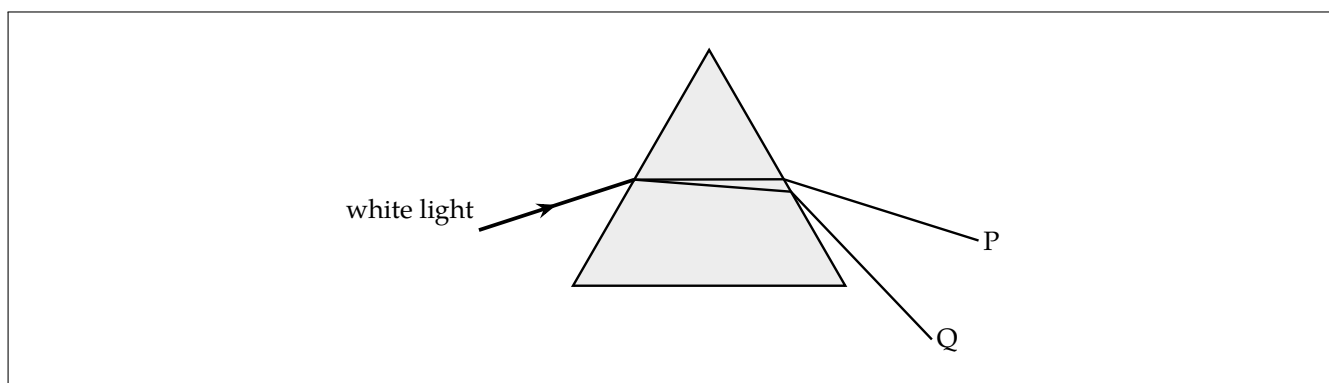
C1 A ray of light hits a glass block. Draw the normal at both faces, the refracted ray inside the block, and the ray that leaves the block.



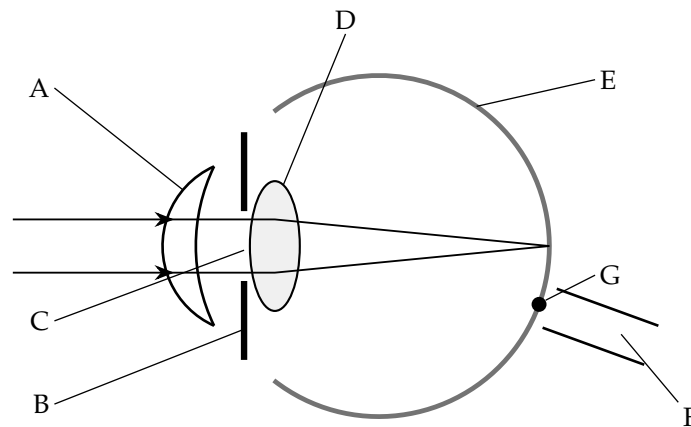
C2 A spear fisher looks at a fish. Draw the ray of light from the fish to the eye, and mark where the fish appears to be.



C3 White light is dispersed by the prism. Which ray, P or Q, is red? Explain.



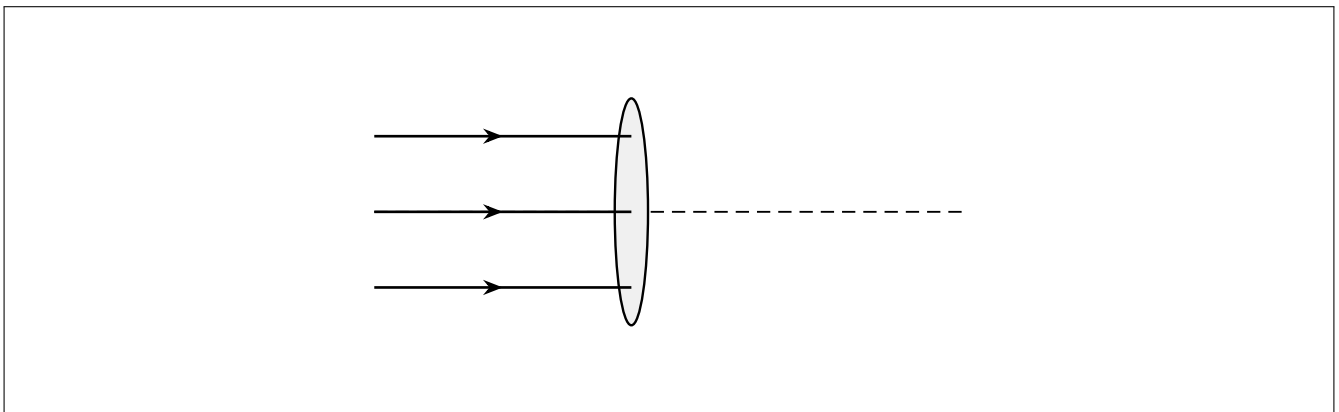
D1 Name the parts of the eye labelled A to G.



A _____ B _____ C _____ D _____

E _____ F _____ G _____

D2 Parallel rays reach a convex lens. Complete the rays after the lens. Mark the focal point F and the focal length.



D3 Why can no image be formed at the blind spot?

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